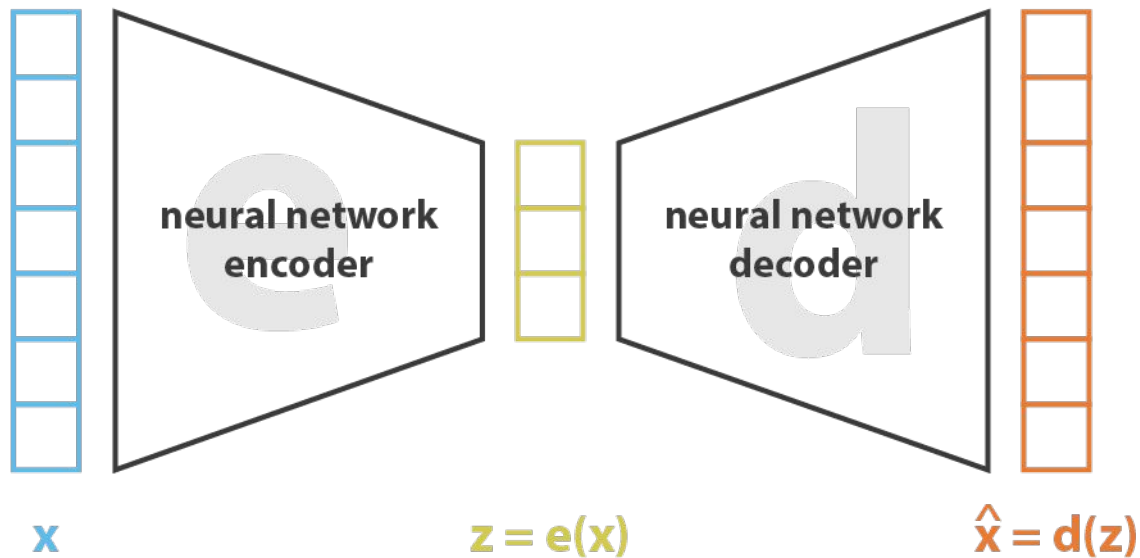




Variational Autoencoders (VAE)

LAB Team

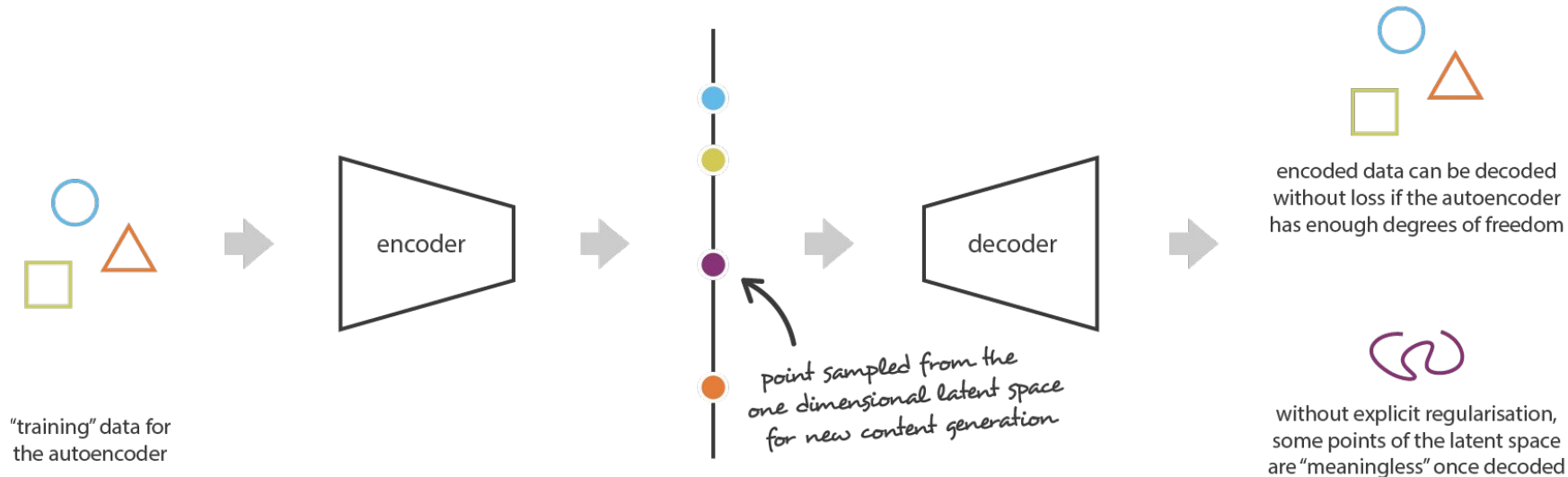
Autoencoders



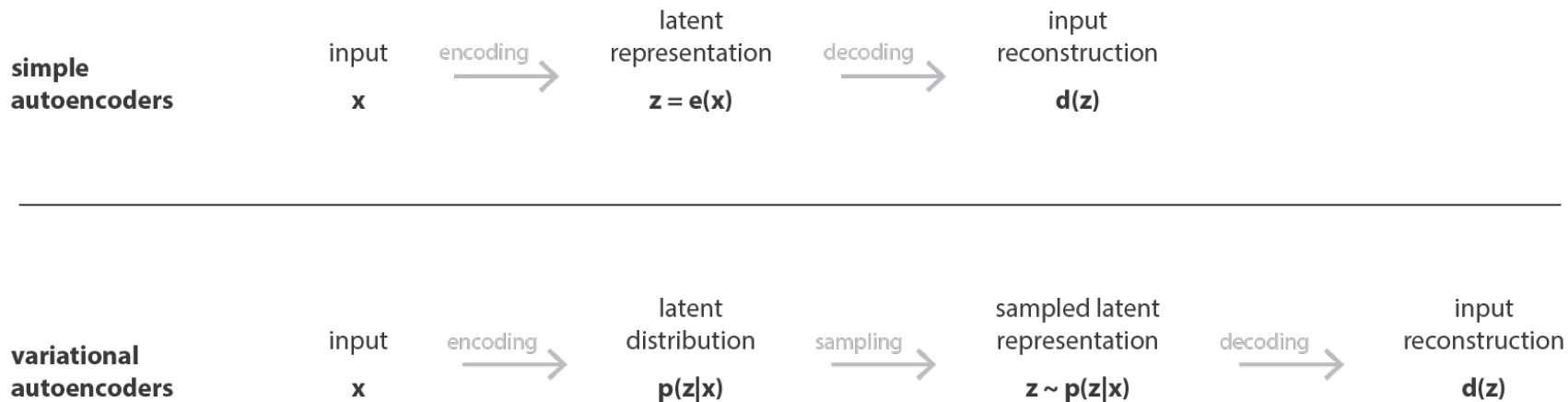
$$\text{loss} = \|x - \hat{x}\|^2 = \|x - d(z)\|^2 = \|x - d(e(x))\|^2$$

Limitations of autoencoders

Generating new content???

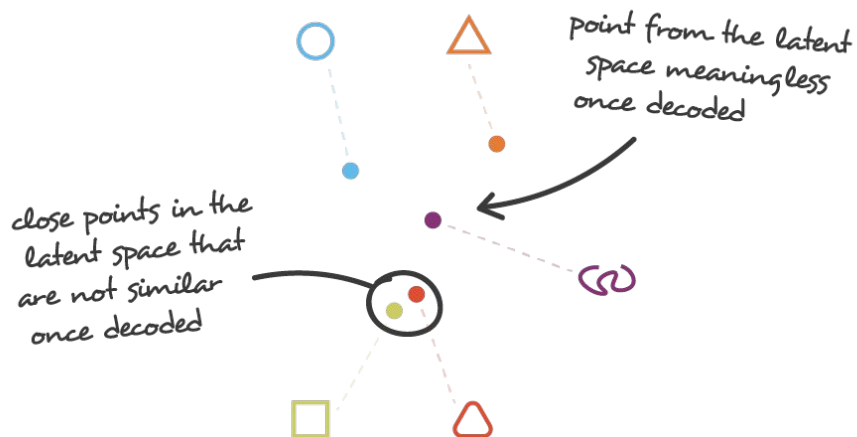


VAE vs Simple autoencoders

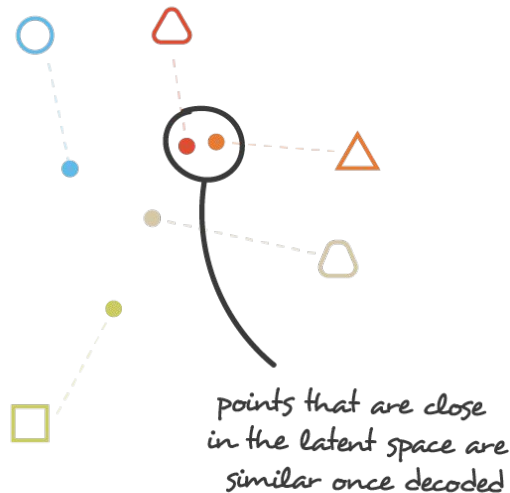


Intuitions about regularization

Continuity and completeness

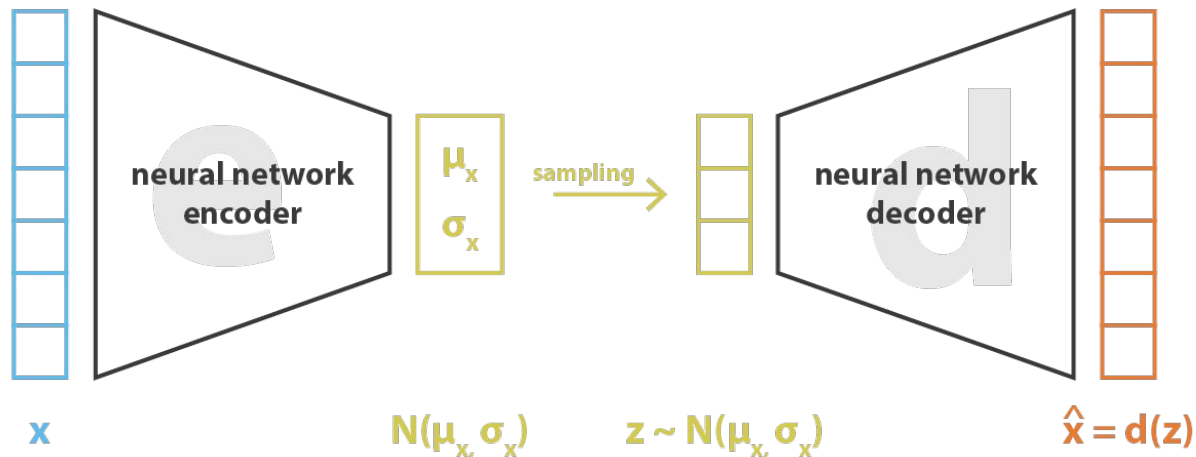


irregular latent space



regular latent space

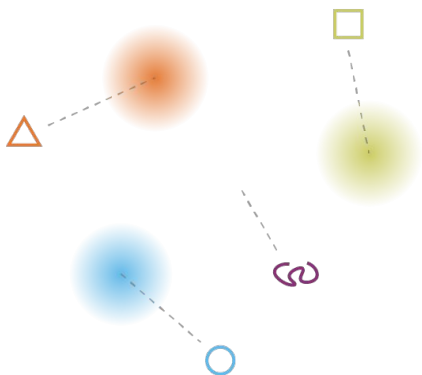
VAE



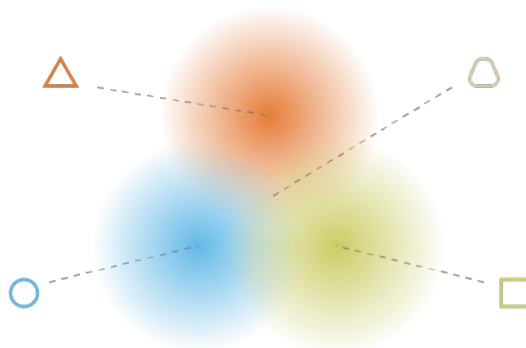
$$\text{loss} = \|x - \hat{x}\|^2 + \text{KL}[N(\mu_x, \sigma_x), N(0, I)] = \|x - d(z)\|^2 + \text{KL}[N(\mu_x, \sigma_x), N(0, I)]$$

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad \text{or} \quad D_{\text{KL}}(P \parallel Q) = - \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{Q(x)}{P(x)} \right)$$

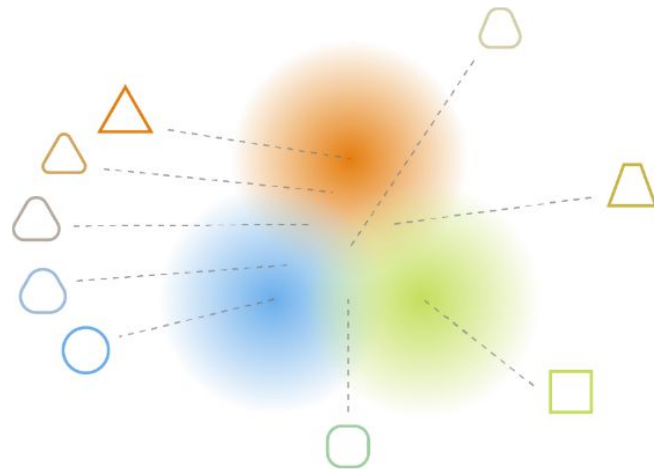
Result of regularization



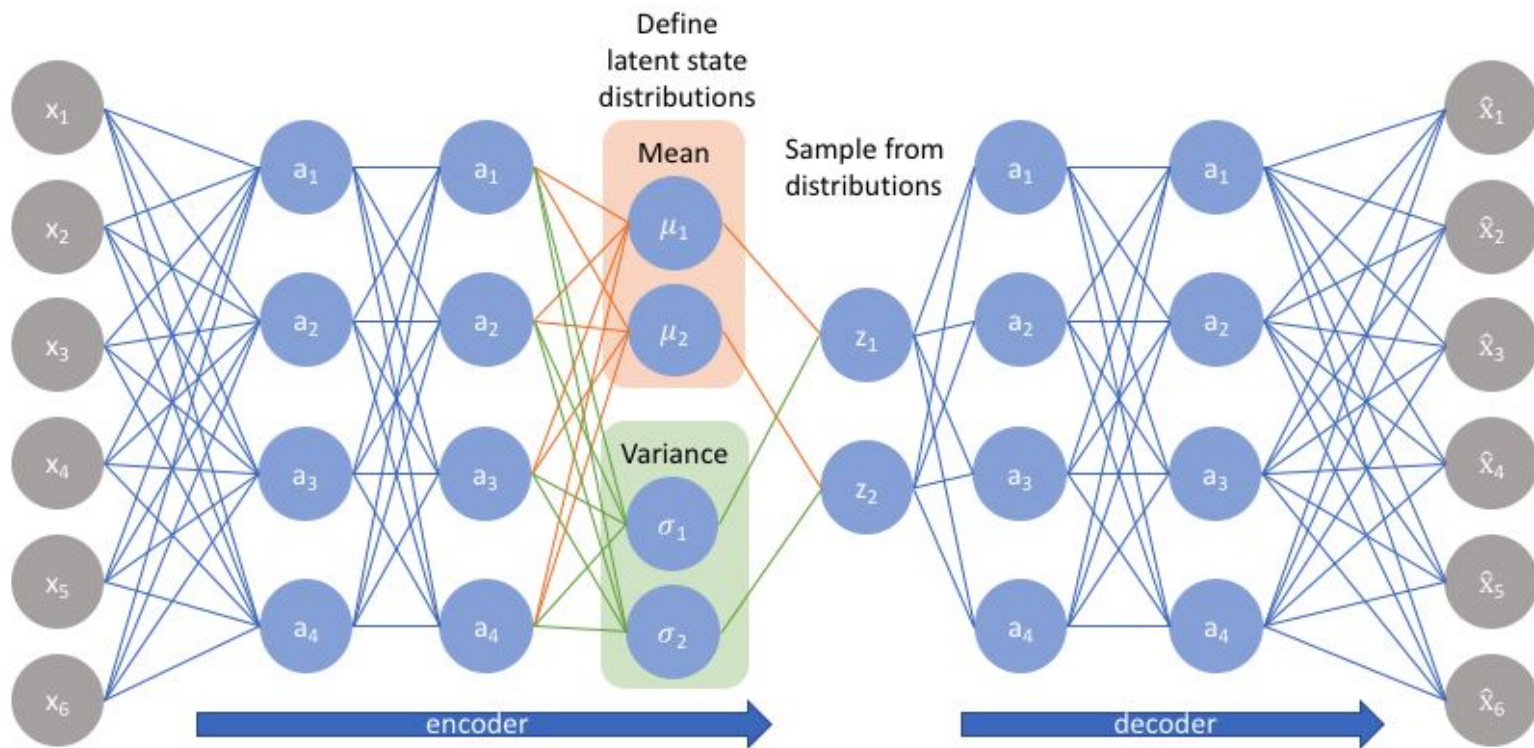
what can happen without regularisation



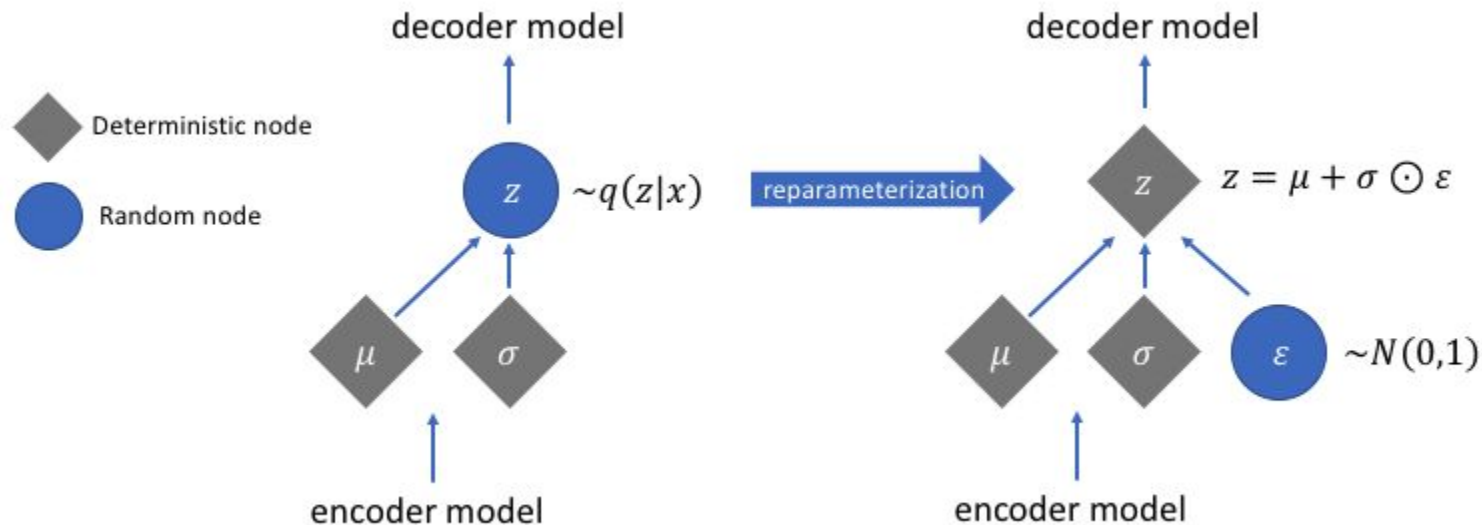
what we want to obtain with regularisation



Detail of VAE



Reparameterization



Backpropagation

